

John Matthew Van

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Education

University of California, Berkeley | GPA: 3.9/4.0

Master of Information and Data Science

May 2024

University of California, Berkeley

Data Science, B.A. & Economics, B.A. (Double Major)

May 2021

Skills & Tools

Programming: Python (pandas, NumPy, SciPy, scikit-learn, statsmodels), SQL (window functions, CTEs), R

Experimentation & Causal Inference: A/B Testing, hypothesis testing, power analysis, CUPED, sequential testing, propensity score matching, difference-in-differences, regression modeling

Machine Learning: XGBoost, CatBoost, LightGBM, PyTorch, HuggingFace, Claude/GPT (fine-tuning & evaluation)

Data Platforms: Google Cloud Platform, BigQuery, Snowflake, Databricks, AWS, Airflow, Git, Tableau, Spark

Work Experience

CVS Health | Woonsocket, RI

Data Scientist

July 2024 – Present

- Led development of a multi-model member-level risk-stratification platform serving 3M+ Medicare members, generating \$8M+ in revenue through reduced leakage and improved case mix
- Built & deployed XGBoost & CatBoost health risk models for readmission and new-member risk prediction, achieving an AUC of 0.76; scores power daily clinical outreach decisions
- Conducted comparative analysis revealing the IP6 model outperforms New Member risk model after month 2, contradicting core modeling assumptions; presented findings to VP of Clinical Analytics, driving org-wide model adoption, increasing medical cost savings by \$2M/year
- Designed and shipped an automated validation pipeline for the Medicaid risk product, aggregating 50+ rule-based and statistical checks (drift, score stability, z-score outliers) to gate production deployments and eliminate manual QA
- Architected and built a receptivity model for Medicaid member engagement; validated via randomized A/B test (treatment vs. control) showing statistically significant 14% lift in response rate

CVS Health | Woonsocket, RI

Data Science & Analytics Intern

June 2023 – Aug 2023

- Built a multi-linear regression model in Python to score customer retention risk, achieving an R-squared of 0.82
- Queried & engineered features from 300M+ rows of transactional data via SQL, identifying 25 final key features

Infosys Limited | Los Angeles, CA

Software Engineer

June 2021 – Jan 2023

- Architected Python & Tableau analytics dashboards, enabling KPI visibility for 4 distinct enterprise clients
- Led end-to-end migration of a ~25TB legacy SQL database to Snowflake, cutting infrastructure costs by \$50k/year

Projects

Fraudulent Job Postings

- Fine-tuned BERT, Bag of Words, TF-IDF, & GloVe embeddings for fake job detection; TF-IDF achieved F1 of 0.91
- Resolved class imbalance via dataset rebalancing/BERT class-weight tuning, optimizing for precision-recall tradeoff

DeeBoT

- Engineered GPT-4 into a personalized DBT therapy chatbot through custom system prompt design, synthetic conversation generation and domain-specific fine-tuning on core DBT skills
- Designed & deployed a rigorous evaluation framework scoring 81 synthetic conversations across 23 DBT adherence criteria, executing 9,315 GPT-4 evaluation calls and achieving 70% DBT standard adherence

Interests

Swimming | Junior Olympian, Swim Captain

- Ranked Top 30 in California high school swim, qualifying to the 2017 State Championship in Fresno, CA